

RailSync AI - Software Requirements Specification (SRS)

Problem Statement Analysis

Manual maintenance block planning is fragmented across Engineering, TRD and S&T.; RailSync centralizes planning and optimization.

Solution Overview

Integrated Railway Planning State + Maintenance Intelligence + Corridor Availability + CP-SAT Optimization.

AI & ML

XGBoost/Scikit-Learn for risk and priority scoring. SHAP for explainability. OR-Tools CP-SAT for scheduling optimization.

Architecture

Data Sources -> Integration -> Planning State -> Opportunity Engine -> Resource Engine -> Optimizer -> Validator -> Approval.

Technology Stack

Next.js, FastAPI, PostgreSQL, XGBoost, SHAP, OR-Tools, Tailwind, Recharts.

Functional Requirements

Maintenance requests, railway network, train occupancy, optimization, approval workflow, KPI engine, dynamic replanning.

Roadmap

16 milestones from maintenance request system to production-ready command center.